

ASPIRE COLLEGE VEHARI CAMPUS

Notes • BS CS 3rd Semester • Lecture 14: Network Layer & Routing Concepts

Course: Computer Networks (Credit Hours: 3)

Instructor: Muhammad Usman Ghani

1. WHAT IS THE NETWORK LAYER? (LAYER 3)

While the Data Link Layer is only responsible for delivery between two adjacent devices on the same wire, the **Network Layer** is responsible for **End-to-End host delivery** across multiple intermediate networks and routers around the world.

Two Main Jobs of Network Layer:

- **Logical Addressing:** Adding global IP addresses (Source IP and Destination IP) to create a **Packet**.
- **Routing:** Determining the optimal, lowest-cost path from source network to destination network.

2. ROUTING VS. FORWARDING

Routing (Finding the Path)

The network-wide process of calculating and selecting the best end-to-end paths across routers.

- Builds the **Routing Table**.
- Operates on the **Control Plane**.
- **Everyday Analogy** Looking at Google Maps to plan your entire trip from Vehari to Lahore.

Forwarding (Switching the Packet)

The local action taken by a single router when a packet enters one port and moves to the exit port.

- Consults the Routing Table for match.
- Operates on the **Data Plane**.
- **Everyday Analogy** Reaching a roundabout and turning into the specific road towards the motorway.

3. TYPES OF ROUTING: STATIC VS. DYNAMIC

Feature	Static Routing	Dynamic Routing
Table Configuration	Routes are typed manually by the network administrator.	Routers exchange info and learn routes automatically via protocols.
Reaction to Link Failure	Cannot adapt; traffic drops until admin manually updates the route.	Reroutes traffic automatically within seconds if a link fails.
Resource Overhead	Zero CPU/memory overhead; no bandwidth wasted on route advertisements.	Consumes router CPU and periodically uses link bandwidth for updates.
Best Used For	Small, simple networks (e.g., small branch office with 1 router).	Large enterprise networks and ISP / Internet backbones.

4. OVERVIEW OF DYNAMIC ROUTING PROTOCOLS

Dynamic routing algorithms calculate optimal paths using metric costs (Hop count, Delay, Bandwidth):

- **Distance Vector (e.g., RIP):** Routers share their entire routing table only with immediate neighbors. Uses **Hop Count** as metric. (Concept: "Routing by rumor").
- **Link State (e.g., OSPF):** Every router advertises its link states to the entire network, building a complete network topology map using **Dijkstra's Algorithm**. Faster convergence.

5. QUICK STUDENT REVIEW QUESTIONS

Q1: What is the main difference between a MAC address and an IP address?

Ans: A MAC address is a permanent hardware identity used only inside a local LAN, whereas an IP address is a logical location address used to route data across different networks over the Internet.

Q2: What is meant by "Hop Count" in routing?

Ans: A Hop Count is the number of intermediate routers a packet must pass through to reach its final destination network.